Lecture VI - Minimizing Split Orders in E-Commerce

Applied Optimization with Julia

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Introduction

E-Commerce Trends

Question: What are current trends in e-commerce?

E-Commerce Sales

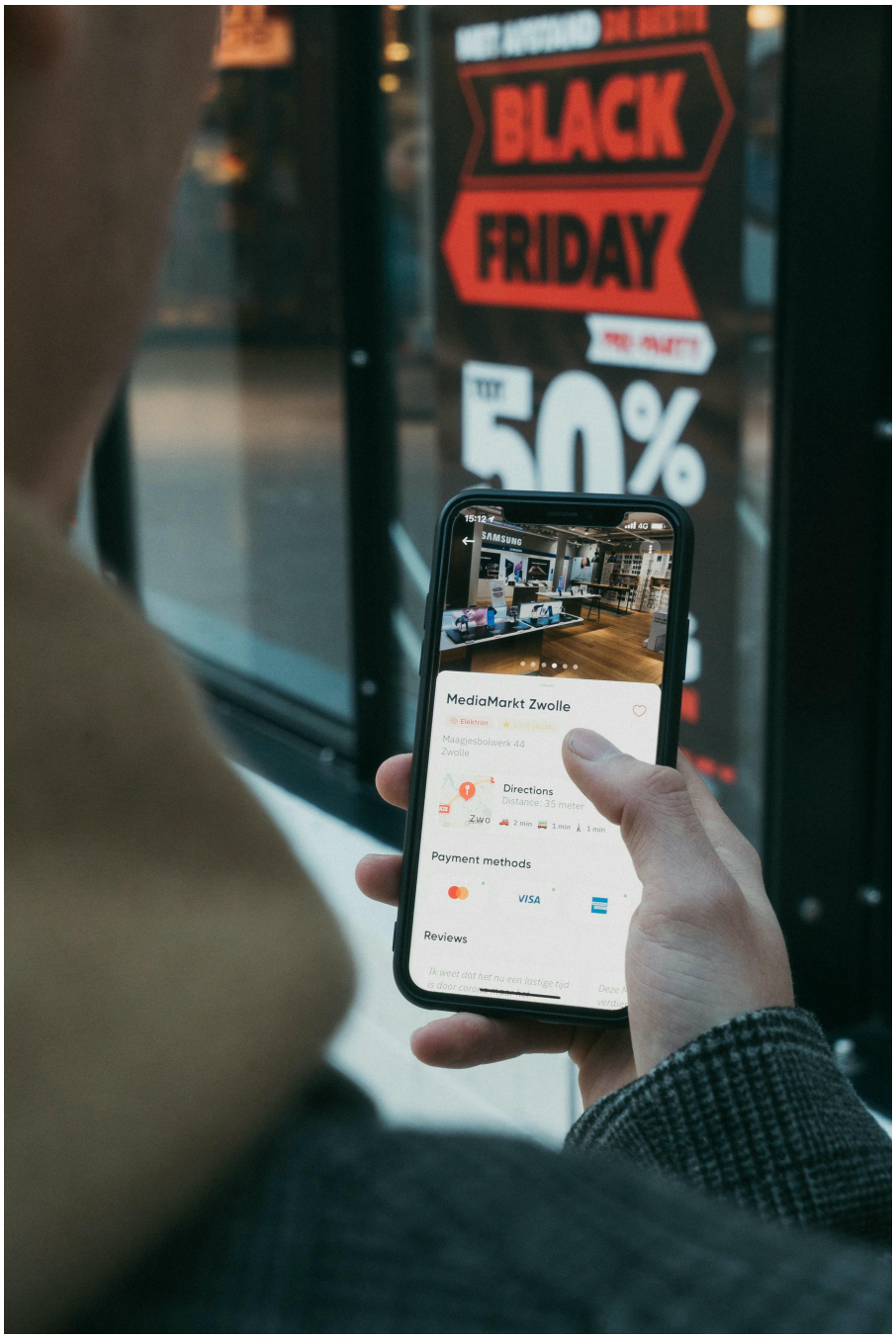
- E-Commerce sales are growing fast:
  - Products are **no longer bound between borders**
  - Product variety is **rising**
  - Consumer shopping patterns are **shifting**
  - Brick-and-mortar stores **loose customers to the internet**
  - Covid-19 **accelerated this trend even more**

Parcels Worldwide

- The number of parcels is rising:
  - **2014**: 44 billion parcels [1]
  - **2019**: 103 billion parcels [2]
  - **2026**: 220 – 262 billion parcels[1] [3]

Pressure on infrastructure

[1]Forecast, not actual number



- Consumers nowadays expect free and fast deliveries and returns
- Existing warehouses have to store an **increasing range of products**
- Better customer service requires **faster deliveries**
- Incurred fulfillment costs **depend on the number of parcels**

Pressure on the environment

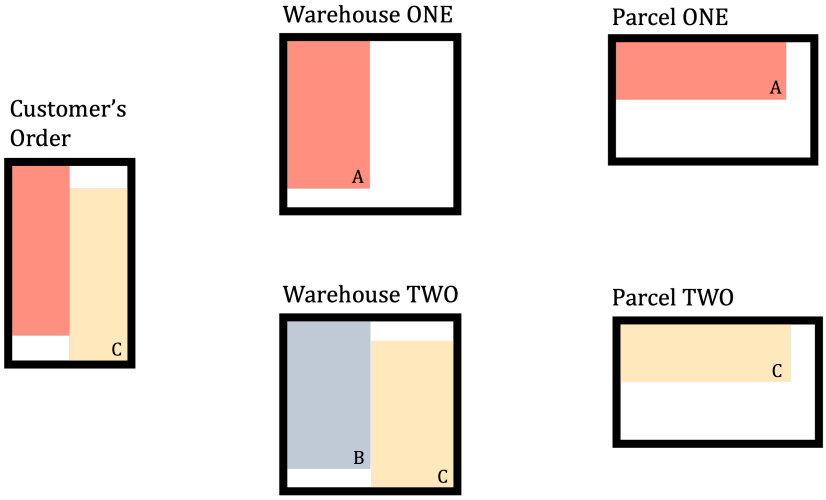
- **Each parcel packaging** consumes resources during production
- Every dispatched parcel to the customer causes CO_2 emissions
- In case of returns, **more parcels cause more emissions**

Problem Structure

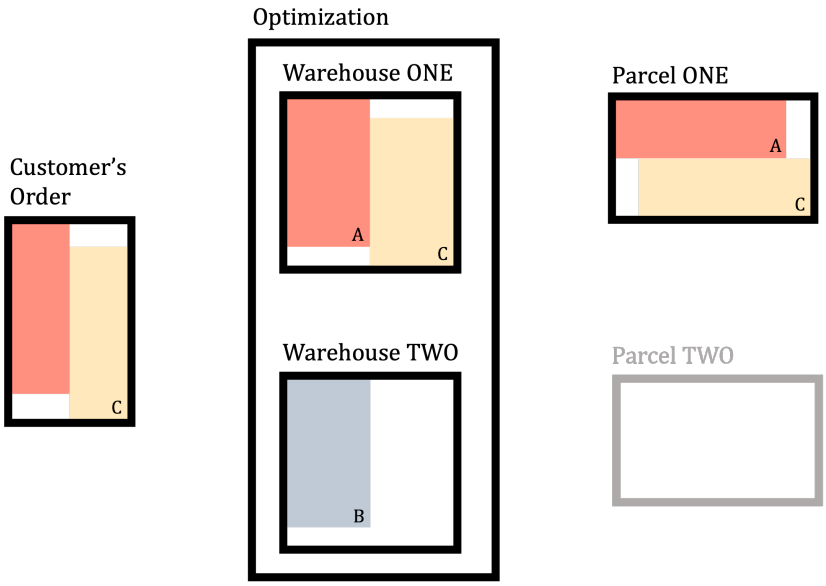
Split Order

Question: **What is a split order?**

. . .



No Split Order



Reason for Split Orders

Question: **Why might they occur?**

. . .

- **Stock availability**: Some products are out of stock at a warehouse and need to be fulfilled from another warehouse
- **Capacity constraints**: Some products are stored at different warehouses and need to be shipped from elsewhere

Impact of Split Orders

Question: **What are the consequences?**

. . .

- **Higher** shipping costs

- **Increased** packaging material
- **More** CO_2 emissions
- Higher operational complexity
- **Lower** customer satisfaction

Mitigations?

Question: **What are possible mitigations?**

. . .

- **Consolidation**: Ship to a central warehouse before dispatch
- **Cross-docking**: Ship directly from supplier to customer
- **Transshipment**: Ship between warehouses before delivery
- **Co-allocation**: Predict co-appearance of products and allocate them to the same warehouse

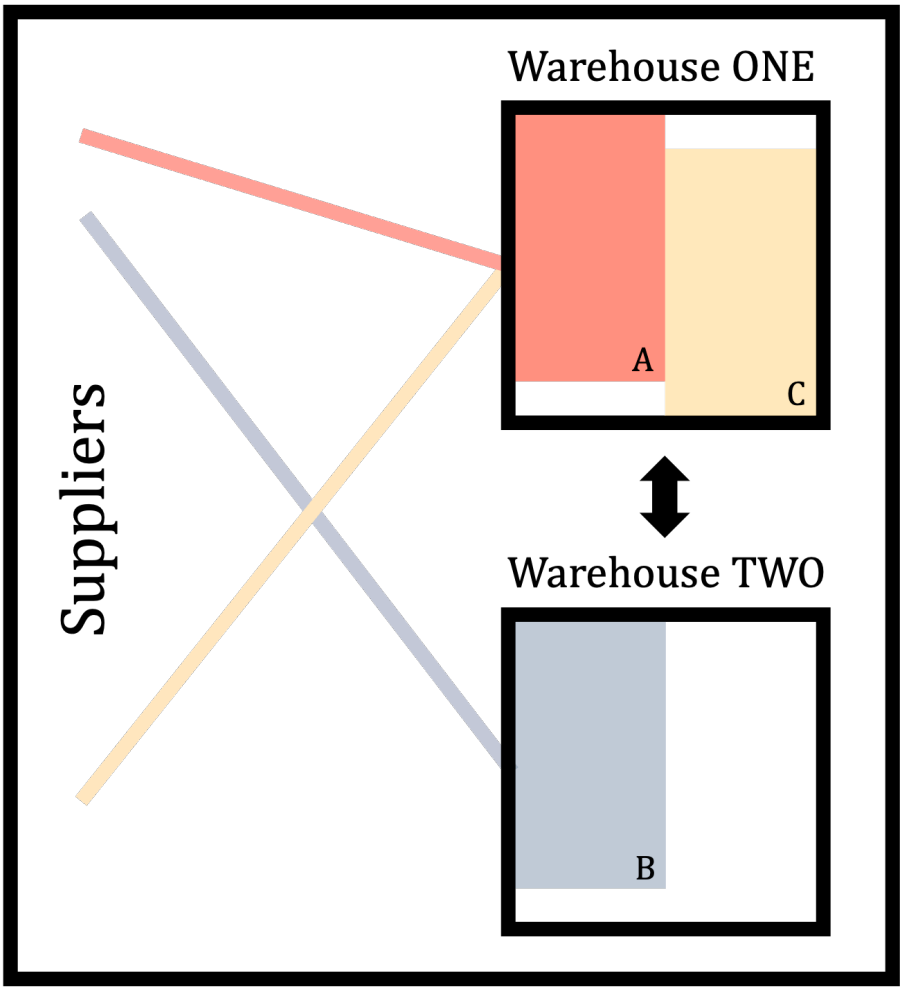
Case Study

Key information about the case:

- a large European e-commerce retailer
- the retailer has two warehouses
- product range cannot be stored in either warehouse
- product deliveries can be made to both warehouses
- products do not have to be stored exclusively

Problem Structure - Version 1

Optimizing Co-allocation



Question: What could be our objective?

We aim to improve the **SKU[2]-warehouse allocation** to minimize the number of split parcels resulting from **SKUs being stored in different warehouses**.

Available Sets

Question: **What could be the sets here?**

. . .

- $\mathcal{I}$ - Set of products indexed by $i \in \{1, 2, ..., |\mathcal{I}|\}$
- $\mathcal{K}$ - Set of warehouses indexed by $k \in \{1, ..., |\mathcal{K}|\}$
- $\mathcal{M}$ - Set of customer orders $m \in \{1, 2, ..., |\mathcal{M}|\}$

Available Parameters

Question: **What are possible parameters?**

. . .

- c_k - Storage space of warehouse $k \in \{1, ..., |\mathcal{K}|\}$
- $\boldsymbol{T} = (t_{m,i})$ - Past customer orders for SKUs

[2]SKU: Stock Keeping Unit

. . .

Question: What could the transactional data look like?

Transactional Data

$t_{m,i}$	A	B	C	D
1	1	1	1	0
2	1	1	1	0
3	1	1	0	0
4	1	0	0	1
5	1	0	0	1
6	1	0	0	1
7	1	0	0	1
8	0	0	1	1

Past vs. Future

- The transactional data $\boldsymbol{T}$ is based on **past orders**
- It is a **binary matrix** of customer orders and SKUs
- We use this data to **assume** future co-occurrence
  - Past co-occurrence predicts future co-occurrence

. . .

Question: **What is your opinion on the assumption?**

Split-Order Minimization

Question: **What could be our decision variable/s?**

. . .

> We have the following sets:
>
> - $\mathcal{I}$ - Set of products indexed by $i \in \{1, 2, ..., |\mathcal{I}|\}$
> - $\mathcal{K}$ - Set of warehouses indexed by $k \in \{1, ..., |\mathcal{K}|\}$
> - $\mathcal{M}$ - Set of customer orders $m \in \{1, 2, ..., |\mathcal{M}|\}$

. . .

- $X_{i,k}$ - 1, if $i \in \mathcal{I}$ is stored in $k \in \mathcal{K}$, 0 otherwise
- $Y_{m,i,k}$ - 1, if SKU $i \in \mathcal{I}$ is shipped from warehouse $k \in \mathcal{K}$ for customer order $m \in \mathcal{M}$, 0 otherwise

Integer Programming Model

- A. Catalán and M. Fisher [4] created an **integer model**
- Number of SKUs of E-Commerce retailers can easily be **between 10,000 - 100,000**

- Number of customer orders necessary for "stable" results have to be higher in the order of **100,000 - 10,000,000**

. . .

Question: **Anybody an idea what this could mean?**

Implementation Challenges

- Small instance with 10 SKUs and 1000 customer orders
- **CPLEX 20.1.0** needs 3100 seconds to solve the problem
- Computation times scales exponentially
- → **Not applicable** in real world applications!

Any idea what

could be done?

Problem Structure - Version 2

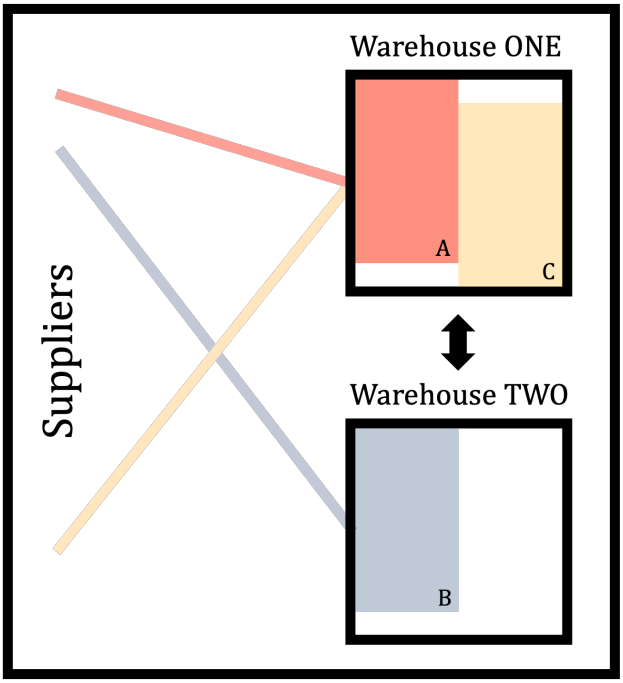
Heuristic Approach

- **Heuristic**: Fast, but not necessarily optimal
- **Approximation**: Not guaranteed to be optimal, but close
- **Computational Effort**: Reasonable even for large instances

. . .

> Different view on the problem
>
> Focus on the warehouses and the co-appearance of SKUs! Discard the exact information about the customer orders.

Objective



Question: **What could be the objective?**

Maximize the coappearance of products that are often **part of the same customer orders**.

Transaction Matrix

```
T = [
    1 1 1 0;
    1 1 1 0;
    1 1 0 0;
    1 0 0 1;
    1 0 0 1;
    1 0 0 1;
    1 0 0 1;
    0 0 1 1
]

# Create the coappearance matrix
Q = T' * T
println("Coappearance matrix Q:")
display(Q)
```

```
Coappearance matrix Q:
```

```
4×4 Matrix{Int64}:
 7  3  2  4
 3  3  2  0
 2  2  3  1
 4  0  1  5
```

Coappearance Matrix

- $\boldsymbol{Q}$ is a symmetric matrix
- Proposed by A. Catalán and M. Fisher [4]
- $\boldsymbol{Q} = (\boldsymbol{T}^T \cdot \boldsymbol{T})$ where $\boldsymbol{Q} = (q_{ij})_{i \in \{1,\ldots,\mathcal{I}\}, j \in \{1,\ldots,\mathcal{I}\}}$
- q_{ij} shows how often i and j appear **in the same order**

. . .

Question: **What do the principal diagonal values tell us?**

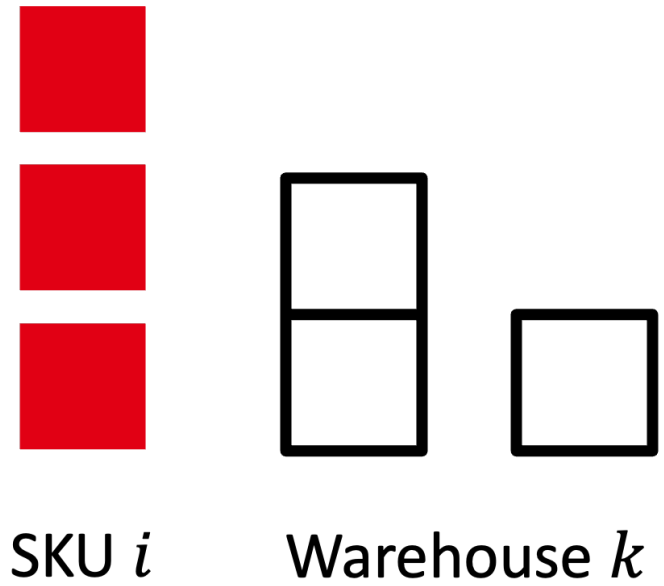
. . .

- How often each SKU appeared over all orders **(binary!)**

How to approach the problem?

- **Greedy Heuristic**[3]: Allocation based on matrix
- **Mathematical Model**[4]: Maximizes coappearance
- **GRASP**[5]: Good on small instances
- New: Max. coappearance with non-linear solver
- New: Heuristic based on Chi-Square Tests

Basic Setting



Available Data (Version 2)

Question: **What could be the sets?**

. . .

- $\mathcal{I}$ - Set of products indexed by $i \in \{1, 2, \ldots, |\mathcal{I}|\}$

[3] Simple and very fast, A. Catalán and M. Fisher [4]
[4] Computationally intensive with CPLEX, S. Zhu, X. Hu, K. Huang, and Y. Yuan [5]
[5] Greedy Randomized Adaptive Search Procedure, S. Zhu, X. Hu, K. Huang, and Y. Yuan [5]

- $\mathcal{K}$ - Set of warehouses indexed by $k \in \{1, ..., |\mathcal{K}|\}$

. . .

! No customer order information is needed!

We can focus on the SKUs and the warehouses, making the problem **much smaller**!

Available Parameters

Question: **What are possible parameters?**

- c_k - Storage space of warehouse $k \in \{1, ..., |\mathcal{K}|\}$
- $\boldsymbol{Q} = \left(q_{ij}\right)_{i \in \{1,...,\mathcal{I}\}, j \in \{1,...,\mathcal{I}\}}$ - Coappearance matrix

. . .

! Transactional Data replaced

Instead of the transactional data, we just **use the coappearance matrix** in our model!

Model Formulation

Decision Variables?

i We have the following sets:

- $\mathcal{I}$ - Set of products indexed by $i \in \{1, 2, ..., |\mathcal{I}|\}$
- $\mathcal{K}$ - Set of warehouses indexed by $k \in \{1, ..., |\mathcal{K}|\}$

. . .

! Our objective is to:

Maximize the coappearance of products that are often part of the same customer orders. **In more mathematical terms:** Maximize the sum of all unique pair-wise values $q_{i,j}$ of all SKUs stored in the same warehouse.

. . .

Question: **What could be our decision variable/s?**

Decision Variables

- $X_{i,k}$ - 1, if SKU $i \in \mathcal{I}$ is stored in $k \in \mathcal{K}$, 0 otherwise

. . .

! Only one variable per SKU and warehouse!

As we don't need the customer order information, we only need to make a decision for each SKU and warehouse pair!

Decision Variable in Julia

Question: **How could we formulate the variable in Julia?**

```
import Pkg; Pkg.add("SCIP")
using JuMP, SCIP # SCIP is a non-commercial MIQCP solver

warehouses = ["Hamburg", "Berlin"] # Add warehouses as a vector
skus = ["Smartphone", "Socks", "Charger"] # Add SKUs as a vector

warehouse_model = Model(SCIP.Optimizer)
```

...

```
@variable(warehouse_model, X[i in skus, k in warehouses], Bin)
```

```
2-dimensional DenseAxisArray{JuMP.VariableRef,2,...} with index sets:
    Dimension 1, ["Smartphone", "Socks", "Charger"]
    Dimension 2, ["Hamburg", "Berlin"]
And data, a 3×2 Matrix{JuMP.VariableRef}:
 X[Smartphone,Hamburg]  X[Smartphone,Berlin]
 X[Socks,Hamburg]       X[Socks,Berlin]
 X[Charger,Hamburg]     X[Charger,Berlin]
```

Objective Function

i We need the following:

- $X_{i,k}$ - 1, if SKU $i \in \mathcal{J}$ is stored in $k \in \mathcal{K}$, 0 otherwise
- q_{ij} - Coappearance of SKU $i \in \mathcal{J}$ and $j \in \mathcal{J}$

! Our objective is to:

Maximize the sum of all unique pair-wise values $q_{i,j}$ of all SKUs stored in the same warehouse. Note, that this is a **quadratic objective function**!

...

Question: **What could the objective function look like?**

...

Quadratic Objective Function

$$\text{maximize} \quad \sum_{i=2}^{\mathcal{J}} \sum_{j=1}^{i-1} \sum_{k \in \mathcal{K}} X_{ik} \times X_{jk} \times q_{ij}$$

...

> ℹ This is a **quadratic objective function**!
>
> The quadratic terms are $X_{ik} \times X_{jk}$. This objective function is based on the **Quadratic Multiple Knapsack Problem (QMKP)**, formulated by A. Hiley and B. A. Julstrom [6].

Objective Function in Julia

Question: **How could we formulate this in Julia?**

...

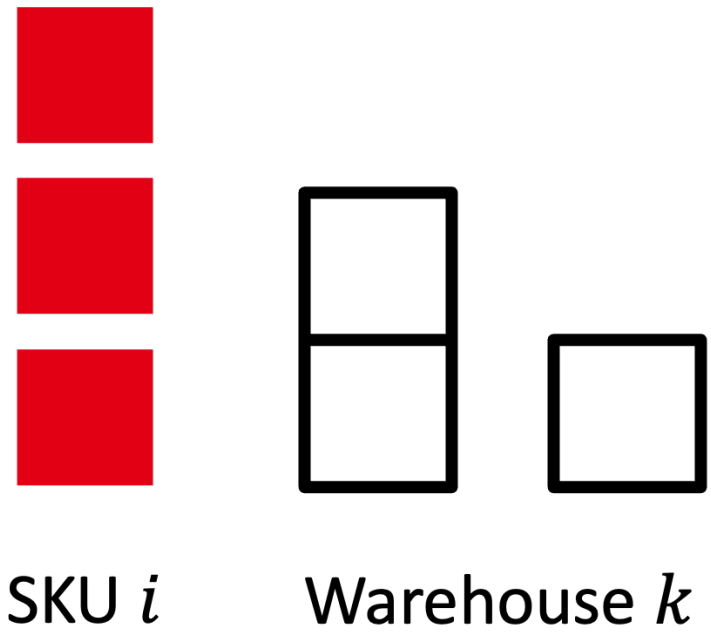
```
Q = [2 1 2; 1 2 1; 2 1 2]

@objective(warehouse_model,
    Max,
    sum(
        X[skus[i], warehouses[k]] * X[skus[j], warehouses[k]] * Q[i,j]
        for i in 2:length(skus)
        for j in 1:i-1
        for k in 1:length(warehouses)
    )
)
```

```
X[Socks,Hamburg]*X[Smartphone,Hamburg] +
X[Socks,Berlin]*X[Smartphone,Berlin] + 2
X[Charger,Hamburg]*X[Smartphone,Hamburg] + 2
X[Charger,Berlin]*X[Smartphone,Berlin] +
X[Charger,Hamburg]*X[Socks,Hamburg] + X[Charger,Berlin]*X[Socks,Berlin]
```

Constraints

What constraints?



Question: **What constraints?**

- Allocate each SKU **at least once**
- Warehouses have a **finite capacity**
- Capacity is **not exceeded**

Single Allocation Constraint?

! The goal of this constraint is to:

Ensure that each SKU is allocated at least once.

. . .

i We need the following variable:

- $X_{i,k}$ - 1, if SKU $i \in \mathcal{I}$ is stored in $k \in \mathcal{K}$, 0 otherwise

. . .

Question: **What could the constraint look like?**

Single Allocation Constraint

$$\sum_{k \in \mathcal{K}} X_{ik} \geq 1 \quad \forall i \in \mathcal{I}$$

. . .

ℹ Remember, this is the variable:

- $X_{i,k}$ - 1, if SKU $i \in \mathcal{I}$ is stored in $k \in \mathcal{K}$, 0 otherwise

. . .

Question: **How could we change the constraint to ensure that each SKU is allocated only once?**

. . .

Question: **How could we add the constraint in Julia?**

Single Allocation in Julia

```
@constraint(warehouse_model, single_allocation[i in skus],
    sum(X[i, k] for k in warehouses) >= 1
)
```

```
1-dimensional DenseAxisArray{JuMP.ConstraintRef{JuMP.Model,
MathOptInterface.ConstraintIndex{MathOptInterface.ScalarAffineFunction{Float64},
MathOptInterface.GreaterThan{Float64}}, JuMP.ScalarShape},1,...} with index
sets:
    Dimension 1, ["Smartphone", "Socks", "Charger"]
And data, a 3-element Vector{JuMP.ConstraintRef{JuMP.Model,
MathOptInterface.ConstraintIndex{MathOptInterface.ScalarAffineFunction{Float64},
MathOptInterface.GreaterThan{Float64}}, JuMP.ScalarShape}}:
 single_allocation[Smartphone] : X[Smartphone,Hamburg] +
X[Smartphone,Berlin] ≥ 1
 single_allocation[Socks] : X[Socks,Hamburg] + X[Socks,Berlin] ≥ 1
 single_allocation[Charger] : X[Charger,Hamburg] + X[Charger,Berlin] ≥ 1
```

Capacity Constraints?

! The goal of these constraints is to:

Ensure that the capacity of each warehouse is not exceeded.

. . .

ℹ We need the following variables and parameters:

- $X_{i,k}$ - 1, if SKU $i \in \mathcal{I}$ is stored in $k \in \mathcal{K}$, 0 otherwise
- c_k - Storage space of warehouse $k \in \mathcal{K}$

. . .

Question: **What could the second constraint be?**

Capacity Constraints

$$\sum_{i \in \mathcal{I}} X_{ik} \leq c_k \quad \forall k \in \mathcal{K}$$

. . .

And that's basically it!

. . .

Question: **How could we add the second constraint in Julia?**

Capacity Constraints in Julia

```
capacities = Dict("Hamburg" => 2, "Berlin" => 1) # Add capacities

@constraint(warehouse_model, capacity[k in warehouses],
    sum(X[i, k] for i in skus) <= capacities[k]
)
```

```
1-dimensional DenseAxisArray{JuMP.ConstraintRef{JuMP.Model,
MathOptInterface.ConstraintIndex{MathOptInterface.ScalarAffineFunction{Float64},
MathOptInterface.LessThan{Float64}}, JuMP.ScalarShape},1,...} with index
sets:
    Dimension 1, ["Hamburg", "Berlin"]
And data, a 2-element Vector{JuMP.ConstraintRef{JuMP.Model,
MathOptInterface.ConstraintIndex{MathOptInterface.ScalarAffineFunction{Float64},
MathOptInterface.LessThan{Float64}}, JuMP.ScalarShape}}:
 capacity[Hamburg] : X[Smartphone,Hamburg] + X[Socks,Hamburg] +
X[Charger,Hamburg] ≤ 2
 capacity[Berlin] : X[Smartphone,Berlin] + X[Socks,Berlin] +
X[Charger,Berlin] ≤ 1
```

QMK Model

$$\text{maximize} \quad \sum_{i=2}^{\mathcal{I}} \sum_{j=1}^{i-1} \sum_{k \in \mathcal{K}} X_{ik} \times X_{jk} \times q_{ij}$$

subject to:

$$\sum_{k \in \mathcal{K}} X_{ik} \geq 1 \; \forall i \in \mathcal{I}$$
$$\sum_{i \in \mathcal{I}} X_{ik} \leq c_k \forall k \in \mathcal{K}$$
$$X_{ik} \in \{0, 1\} \; \forall i \in \mathcal{I}, \forall k \in \mathcal{K}$$

QMK Model in Julia

```
set_attribute(warehouse_model, "display/verblevel", 0) # Hide solver output
optimize!(warehouse_model)
```

```
println("The optimal objective value is: ",
objective_value(warehouse_model))
println("The optimal solution is: ", value.(X))
```

```
The optimal objective value is: 2.0
The optimal solution is: 2-dimensional DenseAxisArray{Float64,2,...} with
index sets:
    Dimension 1, ["Smartphone", "Socks", "Charger"]
    Dimension 2, ["Hamburg", "Berlin"]
And data, a 3×2 Matrix{Float64}:
  1.0  0.0
 -0.0  1.0
  1.0  0.0
```

Model Characteristics

Characteristics

- Is the model formulation **linear/ non-linear?**
- What kind of **variable domain** do we have?
- Do we know the **split-orders** based on the **objective value?**
- Why **couldn't we use HiGHS** as solver?

Choosing a solver

- Identify **problem structure**, e.g. LP, MIP, NLP, QCP, MIQCP, ...
- What is the **size** of the problem?
- Is a **commercial** solver needed?

. . .

> ℹ Commercial Solvers
>
> Commercial solvers are **faster** and **more robust** as open source solvers but also **more expensive**. During your studies, you can use most of them for free though! Nonetheless, we will only use open source solvers in this course.

Global vs Local Optimality

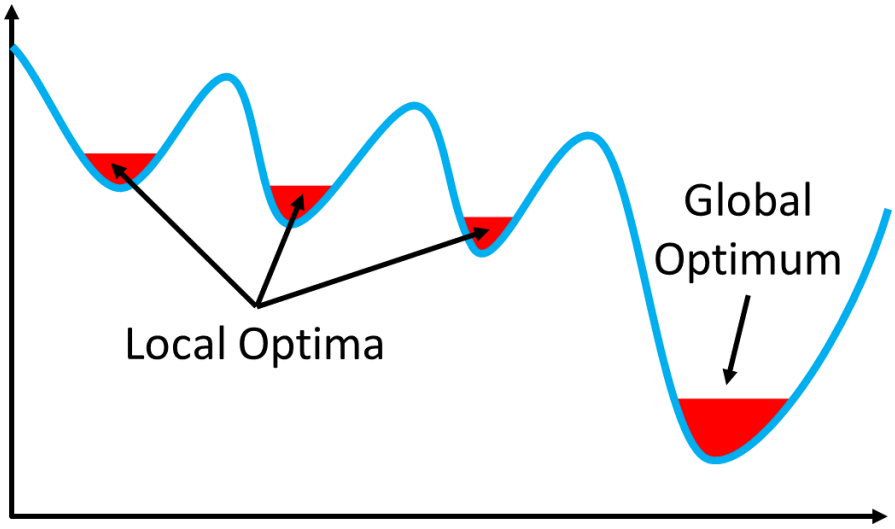


Figure 1: Local vs Global Optimum by Christoph Roser

Model Assumptions

Questions: **On model assumptions**

- What assumptions have we made?
- Problem with allocating SKUs to multiple warehouses?
- What else might pose a problem in the real world?

Impact

Can this be

applied?

Problem Size is Crucial

- Up to 10,000 SKUs ⟶ **commercial solvers**
- More than 10,000 SKUs ⟶ **heuristics**
- For example, the CHI heuristic

. . .

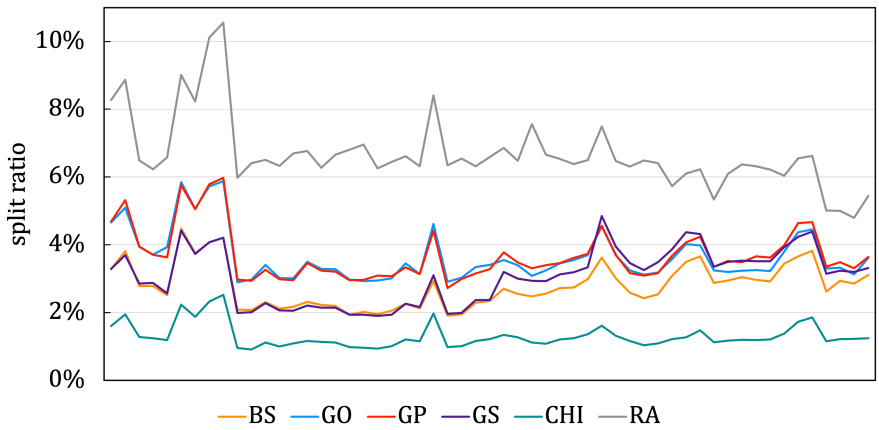
> **CHI-Heuristic**
>
> Detect dependencies between products and allocate them accordingly, as products within orders can have dependencies and products are bought with different frequencies!

Case Study

- More than 100,000 SKUs and several millions of orders
- Comparison of **different heuristics**[6]

- **CHI**: based on Chi-Square tests T. Vlćek and G. Voigt [7]
- **GP, GO, GS, BS**: based on greedy algorithms [4]
- **RA**: Random allocation of SKUs to warehouses

Real Data Set



Conclusion

- Splits are **of no benefit**, except **faster customer deliveries**
- Increase workload, packaging and shipping costs
- Mathematical Optimisation of **"full" problem not solvable**
- **CHI** Heuristic close to mathematical optimisation

> ℹ And that's it for todays lecture!
>
> We now have covered the Quadratic Multiple Knapsack Problem and are ready to start solving some tasks in the upcoming tutorial.

Questions?

Literature

Literature I

For more interesting literature to learn more about Julia, take a look at the literature list of this course.

Bibliography

[1] Pitney Bowes Inc., "Pitney Bowes Parcel Shipping Index Reveals 48 Percent Growth in Parcel Volume since 2014." Accessed: May 27, 2021. [Online]. Available: https://www.businesswire.com/news/home/20170830005628/en/Pitney-Bowes-Parcel-Shipping-Index-Reveals-48

[6] QMKP is not applicable for instance in case study

[2] Pitney Bowes Inc., "Pitney Bowes Parcel Shipping Index Reports Continued Growth Bolstered by China and Emerging Markets." Accessed: May 27, 2021. [Online]. Available: https://www.businesswire.com/news/home/20191010005148/en/

[3] Pitney Bowes Inc., "Pitney Bowes Parcel Shipping Index Reports Continued Growth as Global Parcel Volume Exceeds 100 billion for First Time Ever." Accessed: May 27, 2021. [Online]. Available: https://www.businesswire.com/news/home/20201012005150/en/

[4] A. Catalán and M. Fisher, "Assortment Allocation to Distribution Centers to Minimize Split Customer Orders," *SSRN Electronic Journal*, vol. 0, no. , p. , 2012, doi: 10.2139/ssrn.2166687.

[5] S. Zhu, X. Hu, K. Huang, and Y. Yuan, "Optimization of product category allocation in multiple warehouses to minimize splitting of online supermarket customer orders," *European Journal of Operational Research*, vol. 290, no. 2, pp. 556–571, 2021, doi: 10.1016/j.ejor.2020.08.024.

[6] A. Hiley and B. A. Julstrom, "The Quadratic Multiple Knapsack Problem and Three Heuristic Approaches to It," in *Proceedings of the 8th Annual Conference on Genetic and Evolutionary Computation*, M. Keijzer, Ed., New York, NY: Association for Computing Machinery, 2006, pp. 547–552. doi: 10.1145/1143997.1144096.

[7] T. Vlćek and G. Voigt, "Optimizing SKU-Warehouse Allocations to minimize Split Parcels in E-Commerce Environments," *Submitted to Decision Sciences*, 2024.